

A 67.92% Bound for Simple Critical Zeros in a Vanishing Vertical Box

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Abstract

Let $N(T, 2T)$ count the nontrivial zeros $\rho = \beta + i\gamma$ of the Riemann zeta function with $T < \gamma \leq 2T$, including multiplicity, and let $N_{0,s}(T, 2T)$ count those zeros that are both simple and on the critical line. Goldston and Suriajaya recently proved that the vanishing-box hypothesis

$$\max_{T < \gamma \leq 2T} |\beta - \tfrac{1}{2}| \log T \longrightarrow 0 \quad (1)$$

implies $N_{0,s}(T, 2T) \geq (2/3 - o(1))N(T, 2T)$. We prove under the same hypothesis that

$$N_{0,s}(T, 2T) \geq (0.6792 - o(1))N(T, 2T). \quad (2)$$

The argument combines the unconditional complex-difference Montgomery theorem of Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh with the interval-certified semidefinite test of Chirre, Goncalves, and de Laat. The key step transfers one fixed integrated kernel, rather than a pointwise pair-correlation formula. Endpoint vanishing, two integrations by parts, and an unconditional close-pair estimate make the total perturbation caused by (1) equal to $o(N(T, 2T))$. The result is conditional on (1); it is not an unconditional improvement and has no implication for the truth of the Riemann hypothesis.

Result attribution. Anthropic Claude, in the Claude Fable research session, identified the possible box/semidefinite splice. OpenAI GPT-5.6 Sol found the integrated-kernel proof below, replacing an unproved pointwise comparison, and performed the primary-source and novelty audits. Anthropic Claude Opus then reciprocally audited the repaired proof and found it sound. Michael Devine originated and directed the broader mathematical research program, coordinated the independent audits, and is the accountable author of this manuscript. The model systems are credited as sources of the mathematical work but are not listed as authors because they cannot approve the manuscript or assume responsibility for it.

1. Introduction

Montgomery's pair-correlation method [6], under the Riemann hypothesis (RH), implies that at least two thirds of the nontrivial zeros of the Riemann zeta function are simple. Montgomery and Taylor raised the proportion to about 0.6725, as recorded by Cheer and Goldston [7]. Chirre, Goncalves, and de Laat (CGdL) later used semidefinite programming and the positivity of the ordinate form factor outside the classically evaluated bandwidth to prove, still under RH,

$$N^*(T) \leq (1.3208 + o(1))N(T), \quad N_s(T) \geq (0.6792 - o(1))N(T). \quad (3)$$

Here N^* is the diagonal pair count, or the sum of squared multiplicities.

Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh (BGSTB) subsequently proved an unconditional version of Montgomery's theorem for a form factor built from full complex zero differences. Their later work showed that a vertical box of width $b(T)/\log T$, with $b(T) \rightarrow 0$, recovers a two-thirds lower bound for zeros that are simultaneously simple and on the critical line. Goldston and Suriajaya gave a direct Fejer-kernel proof of that result.

Very recently, a manuscript of Claude (Anthropic) [8] proved unconditionally that at least two thirds of the zeros are distinct critical-line zeros, with the same bound for simple critical-line zeros and the optimized constant 0.6725, by replacing the RH-dependent reading of the zero side with finite-compression inertia. The present note is complementary: its hypothesis (1) is a strong unproved assumption, but under it the resulting constant 0.6792 exceeds what the bandwidth-one inertia method certifies unconditionally. Since RH forces the maximum in (1) to vanish identically, RH implies (1), and under RH the bound (2) is exactly the CGdL bound (3); the content of this note is that the vanishing box already suffices.

The tempting route to (2) is to show that the BGSTB complex-difference form factor converges pointwise to the ordinate form factor throughout the bandwidth $[-1, 1]$. We do not prove or assume that stronger statement. Instead, the CGdL argument only needs an upper bound for one integrated kernel. For its fixed, smooth, endpoint-vanishing certificate, the vanishing box makes this integrated complex-to-ordinate difference $o(N)$. This is enough to transport the certified constant 1.3208.

2. Form factors and ordinate multiplicity

All sums below run over zeros $\rho = \beta + i\gamma$ and $\rho' = \beta' + i\gamma'$ with $T < \gamma, \gamma' \leq 2T$, counted with multiplicity. Write

$$N = N(T, 2T) \sim \frac{T}{2\pi} \log T, \quad w(v) = \frac{4}{4 + v^2}, \quad W(z) = \frac{4}{4 - z^2}. \quad (4)$$

The ordinate and complex-difference form factors, in unnormalized form, are

$$F_{\text{ord}}(\alpha, T) = \sum_{\rho, \rho'} T^{i\alpha(\gamma - \gamma')} w(\gamma - \gamma'), \quad (5)$$

$$F_*(\alpha, T) = \sum_{\rho, \rho'} T^{\alpha(\rho - \rho')} W(\rho - \rho'). \quad (6)$$

The function F_{ord} is nonnegative for every real α , with no hypothesis on the real parts of the zeros. Indeed, w is positive definite, so (5) is a positive quadratic form in the coefficients $T^{i\alpha\gamma}$.

BGSTB's unconditional Montgomery theorem, in the dyadic-window form used by Goldston and Suriajaya, gives uniformly for $|\alpha| \leq 1$

$$\frac{F_*(\alpha, T)}{N} = (1 + o(1)) T^{-2|\alpha|} \log T + |\alpha| + o(1). \quad (7)$$

For a real ordinate γ , let

$$\tilde{m}_\gamma = \#\{\rho : \text{Im } \rho = \gamma\}, \quad N_{\text{ord}}^* = \sum_{\gamma} \tilde{m}_\gamma^2. \quad (8)$$

If $\tilde{m}_\gamma = 1$, the unique zero at that height is simple and lies on the critical line: otherwise its functional-equation mirror $1 - \bar{\rho}$ would be a second zero at the same height. Therefore

$$N_{0,s}(T, 2T) \geq \#\{\gamma : \tilde{m}_\gamma = 1\} \geq 2N - N_{\text{ord}}^*. \quad (9)$$

3. The fixed CGdL certificate

CGdL construct and rigorously certify, using interval arithmetic, a smooth even polynomial-Gaussian function f with last sign-change radius $R = r(f)$, normalized by

$$f(0) = \hat{f}(0) = 1, \quad \hat{f} \geq 0, \quad f(x) \leq 0 \quad (|x| \geq R), \quad f(\pm R) = 0. \quad (10)$$

Its objective satisfies

$$\mathcal{Z}(f) = R + \frac{2}{R} \int_0^R x f(x) dx \leq 1.3208. \quad (11)$$

Define

$$g(x) = R^{-1} \hat{f}(x/R), \quad \hat{g}(\alpha) = f(R\alpha). \quad (12)$$

Since $g \geq 0$, diagonal extraction gives

$$R^{-1} N_{\text{ord}}^* \leq S_{\text{ord}}, \quad (13)$$

where

$$S_{\text{ord}} = \sum_{\rho, \rho'} g\left(\frac{(\gamma - \gamma') \log T}{2\pi}\right) w(\gamma - \gamma'). \quad (14)$$

Fourier inversion expresses (14) as the integral of $\widehat{g}F_{\text{ord}}$. Because $F_{\text{ord}} \geq 0$ and $\widehat{g}(\alpha) = f(R\alpha) \leq 0$ for $|\alpha| \geq 1$, discarding that outer integral gives

$$S_{\text{ord}} \leq S_{\text{in}} = \sum_{\rho, \rho'} K(i(\gamma - \gamma') \log T) w(\gamma - \gamma'), \quad (15)$$

with the entire function

$$K(z) = \int_{-1}^1 f(R\alpha) e^{\alpha z} d\alpha. \quad (16)$$

We compare (15) with the complex-difference sum

$$S_* = \sum_{\rho, \rho'} K((\rho - \rho') \log T) W(\rho - \rho'). \quad (17)$$

By (6), (7), and (16),

$$\begin{aligned} S_* &= \int_{-1}^1 f(R\alpha) F_*(\alpha, T) d\alpha \\ &= N \left(f(0) + 2 \int_0^1 \alpha f(R\alpha) d\alpha \right) + o(N). \end{aligned} \quad (18)$$

The first term in (18) follows from $T^{-2|\alpha|} \log T \rightarrow \delta_0$ in the distributional sense.

4. Integrated-kernel transfer through the box

Set

$$b_T = 2 \log T \max_{T < \gamma \leq 2T} |\beta - \tfrac{1}{2}|, \quad b_T \rightarrow 0, \quad (19)$$

and for each pair put

$$h = (\beta - \beta') \log T, \quad y = (\gamma - \gamma') \log T, \quad \delta = \beta - \beta', \quad v = \gamma - \gamma'. \quad (20)$$

Then $|h| \leq b_T$ and $|\delta| \leq b_T / \log T$.

Let $\phi(\alpha) = f(R\alpha)$ and $q_h(\alpha) = \phi(\alpha)(e^{h\alpha} - 1)$. Since $q_h(\pm 1) = 0$, two integrations by parts show, for $|y| \geq 1$,

$$\left| \int_{-1}^1 q_h(\alpha) e^{i\alpha y} d\alpha \right| \leq \frac{|q'_h(1)| + |q'_h(-1)| + \|q''_h\|_1}{y^2} \ll_f \frac{|h|e^{|h|}}{y^2}. \quad (21)$$

The boundary term involving $q'_h(\pm 1)$ is present unless $f'(\pm R) = 0$; it is explicitly included in (21) and is harmless because $e^{\pm h} - 1 = O(|h|e^{|h|})$. For $|y| < 1$, the direct integral bound has the same numerator. Consequently

$$|K(h + iy) - K(iy)| \ll_f \frac{|h|e^{|h|}}{1 + y^2}, \quad |K(iy)| \ll_f \frac{1}{1 + y^2}. \quad (22)$$

For $|\delta| \leq 1$, the real part of $4 - (\delta + iv)^2$ is at least $3 + v^2$. Direct subtraction therefore gives

$$|W(\delta + iv)| \ll w(v), \quad |W(\delta + iv) - w(v)| \ll \frac{|\delta|}{1 + v^2}. \quad (23)$$

We also require the normalized close-pair mass estimate

$$\sum_{\rho, \rho'} \frac{1}{1 + ((\gamma - \gamma') \log T)^2} \ll N. \quad (24)$$

For completeness, Goldston and Suriajaya record the unconditional estimate, valid for $0 \leq H \leq T$,

$$\#\{(\rho, \rho') : 0 < \gamma, \gamma' \leq 2T, |\gamma - \gamma'| \leq H\} \ll (1 + H \log T) T \log T. \quad (25)$$

The same bound applies to our dyadic subwindow. Split the pairs into $|\gamma - \gamma'| \log T \leq 1$ and the dyadic shells $2^{j-1} < |\gamma - \gamma'| \log T \leq 2^j$. Equation (25) bounds the shell count by $O((1 + 2^j)N)$; multiplication by $O(2^{-2j})$ and summation proves (24).

Using (22)–(24) term by term in (15) and (17) gives

$$\begin{aligned} |S_* - S_{\text{in}}| &\ll_f \sum_{\rho, \rho'} \frac{b_T e^{b_T}}{1 + y^2} + \sum_{\rho, \rho'} \frac{b_T / \log T}{1 + y^2} \\ &\ll_f b_T e^{b_T} N + \frac{b_T}{\log T} N = o(N). \end{aligned} \quad (26)$$

This is the entire complex-to-ordinate transfer needed for the theorem.

5. Main theorem

Theorem. Suppose that (1) holds. Then

$$\liminf_{T \rightarrow \infty} \frac{N_{0,s}(T, 2T)}{N(T, 2T)} \geq 0.6792. \quad (27)$$

Proof. Combining (13), (15), (18), and (26), and then multiplying by R , gives

$$N_{\text{ord}}^* \leq \left(R + 2R \int_0^1 \alpha f(R\alpha) d\alpha + o(1) \right) N = (\mathcal{Z}(f) + o(1))N. \quad (28)$$

The interval-certified CGdL test (11) implies $N_{\text{ord}}^* \leq (1.3208 + o(1))N$. Equation (9) now yields

$$N_{0,s}(T, 2T) \geq (2 - 1.3208 - o(1))N = (0.6792 - o(1))N.$$

This proves (27). $\square$

6. Scope, comparison, and verification

Goldston and Suriajaya obtain $2/3$ under exactly the asymptotic box hypothesis (1). Equation (27) improves that conditional proportion by about 0.01253, or 1.253 percentage points. It also shows that the CGdL 0.6792 conclusion requires less than RH: the vanishing-box hypothesis permits, as a formal condition, off-line zeros whose scaled horizontal displacements tend to zero, as well as finitely many exceptional zeros.

No unconditional conclusion follows from this paper. In particular, (1) is far beyond current zero-density results. We also do not prove pointwise convergence of (6) to (5), and no such assertion is used. The vanishing of the box width is essential to the transfer: for a box of fixed width $b/\log T$ the perturbation (26) is only $O_b(N)$, the certified constant degrades accordingly, and no improvement over the fixed-width constants of BGSTB is claimed in that regime.

The proof was audited twice against the current primary sources. After the integrated-kernel repair, Anthropic Claude Opus checked each step and independently reproduced the external CGdL optimization numerically as 1.320848, consistent with their rigorous interval-certified 1.3208. The published CGdL certificate, rather than this independent floating-point reproduction, is the rigorous numerical input to (27).

Research provenance and responsibility

The broader investigation was initiated and directed by Michael Devine as part of a program studying whether unusual rearrangements of the Dirichlet eta series might expose new mechanisms relevant to zeta zeros. Anthropic Claude identified that the recently developed box-conditional framework might accept the CGdL semidefinite certificate. Its first argument depended on an unproved pointwise kernel comparison. OpenAI GPT-5.6 Sol found the integrated-kernel proof in Sections 3–4, checked the current primary sources, and performed a targeted novelty search. Anthropic Claude Opus then performed a reciprocal audit, accepted the repair, identified the boundary term now displayed in (21), and independently reproduced the external constant.

These model contributions are disclosed as research provenance, not as human peer review. Devine selected and sustained the research program, coordinated the adversarial checks, decided to circulate the result, and assumes responsibility for the claims and text as the human author.

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Classification and keywords

2020 Mathematics Subject Classification: 11M26 (primary), 11M06 (secondary).

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